

Bank Management System

PREPARED BY:

ANEEQA NADEEM

Terminal Project Report

Project Description:

Project Title: Bank Management System

Language Used: Java

Details:

This project simulates a basic banking system using Java. It allows the user to interact with various banking operations through a command-line interface, it is designed to mimic real-world banking workflows for both customers and administrators. The system includes functionalities such as account management, ATM access, transaction logging, loan application and approval, interest and fee calculation, credit card control, complaint handling, and zakat computation. It shows the use of various programming concepts, such as recursions, arrays, loops, File and exception handling, built-in and user-built methods while incorporating real life logistics of calculations in financial systems.

Task Distributions:

Member name	Project Task Ownership:
Sidra Mehak	• Loan approval and applying (Mortgage, Personal, Housing etc) • Credit Card Controls • Customer's Complaint: File Reading and Writing • Compound Interest Calculations (yearly/monthly basis) • Zakat Automation System • Bank Service Fee Management • File Writing for Bank Transaction History with Time stamps • Secure Admin Access • Admin Menu Flow
Aneeqa Nadeem	• Customer Security Checks • Personal, Savings and Corporate Salary account set ups • New Account Creation System • Reading and Writing of Customer Data File (and updating) • ATM Services(Withdraw, Deposit and Transfer) • Balance Inquiry • Integrated bill payment functionality (Water, Electricity, Gas etc.) • Account Data Management • Customer Menu Flow

Feature Highlights:

1. **Admin and Customer Authentication**
2. **Account Security**
3. **Customer's Account Creation**
4. **Customer Record Management**
5. **Personalized Account Types**
 - Offers:
 - Savings Account
 - Salary Account
 - Personal Account
6. **ATM Services:**
 - Withdraw
 - Transfer
 - Deposit
7. **Loan Supervision System**
 - Handles different loan types:
 - Mortgage Loan
 - Business Expansion Loan
 - Credit Builder Loan
 - Personal Loan
8. **Credit Card Monitoring System**
9. **Account Type Reference & Setup**
10. **Compound Interest Calculations**
11. **Transaction Logging System**
12. **Customer Complaint Management**
13. **Zakat Management System**
14. **Fee & Interest Visibility**

Entities/Modules Working:

- **Files used:**
 - 1) **customers.txt** : To store Customer data

2) **complaints.txt** : To store Customer Complaints

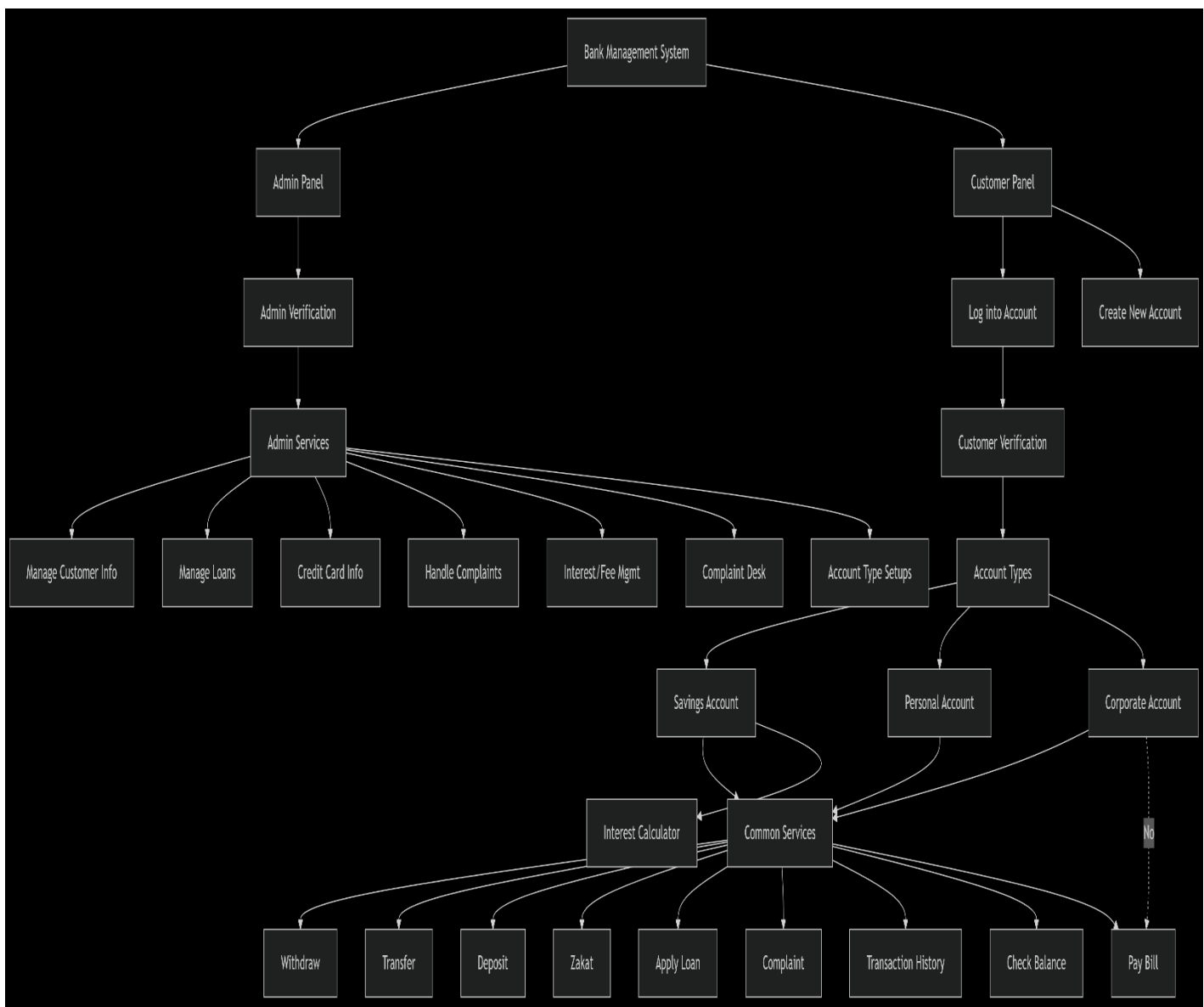
3) **transaction_logs.txt** : To log in Customer's transactions

- **Types of Exceptions being Handled:**

1. InputMismatchException
2. NumberFormatException
3. IllegalArgumentException
4. FileNotFoundException
5. IOException

- **Global Variable:** “**basemoney**” of String type with **value: 1000000000**

- **Global Scanner**



Main Menu:

Entities	Functionalities
1. Customer Portal	• Creates accounts for new users, • Checks customer's authenticity by security checks • provides with all Banking Services
2. Admin Console	• Grants access to Admin Operations Console for managing the Bank System • Requires a secure password for authentication
3. Exit	• Terminates the System and exits the application

Admin Console:

Entities	Functionalities
1. Customer Records Management	Enables the admin to monitor customer profiles efficiently. • View customer details • Maintain organized records. • Provides access to the file customers.txt by file reading
2. Loan Supervision	Handles review and approval of loan applications with multiple loan types. • Business Expansion Loan: For entrepreneurs aiming to scale operations • Mortgage Loan: For purchasing or refinancing property • Credit Builder & Personal Loans: For credit improvement and personal use • Admin can approve or reject requests after basic verification, based on yes/no by using if-else
3. Credit Card Controls	Ensures secure handling of customer credit card operations. • Issue Credit Cards: Assign new cards to eligible customers • Block Cards: Instantly disable compromised or suspicious cards

BMS REPORT

4. Complaint Resolution Desk	Provides access to submitted customer complaints by reading complaints.txt file - Admin reviews complaints submitted by customers Helps identify service flaws and improve customer satisfaction
5. Interest & Fee Manager	Manages interest calculations and applies service fees using compound math. - Uses exponential growth formula to compute accurate interest - Adds a fixed service fee and displays total dues Using mathematical formulae: $A = P \times (1 + r/n)^{nt}$ Use of 2d arrays in calculating for multiple customers
6. Transaction History Review	Allows viewing of all transaction logs stored in a text file. - Displays account activity with amount, type, and timestamp - Useful for verifying complaints or detecting unusual patterns Logs in the real time. - Use of LocalDateTime class from java.util.time package to make real time stamps.
7. Account Type Setups	Displays and manages the interest rates and attributes of the available bank account types. Personal Account. Saving Account. Salary Account.

CUSTOMER CONSOLE:

ENTITIES	FUNCTIONALITIES
1.Customer Panel	- Asks customers to either create a new account - gives customer the facility to log into an existing account
2.Existing Account Login	- verifies if the customer account exists in customer file on the backend or not on the basis of a pin and a specific id - Verification done through method isCredentialsValid - if the account exists, method to display different account types is invoked

BMS REPORT

3.New Account	• Displays various services being provided by the bank • Prompts user to enter their credentials including name, email and type of account • Assigns an ID to user • Invokes add new customer method to update information of new customer to the existing customer file
4.Account Type Menu	• After verification, displays and asks user to choose any one of the account type and invokes their respective methods: • Savings Account • Personal Account • Corporate Salary Account
5.Savings Account	• Displays a user friendly menu with various services • Applies interest by invoking a specific method for it • Invokes corresponding methods for each of the following services: • Check Balance • Withdraw Money • Transfer Money • Deposit Money • Apply for Loan • Zakat Calculation • Pay Bills • File Complaint • View Transaction History • Return to Account type menu
6.Personal Account	• Displays a user friendly menu with various services • Invokes corresponding methods for functions same as savings account method • No Interest applied • Return to Account type menu
7.Corporate Salary Account	• Displays a user friendly menu with various services • Invokes corresponding methods for services same as other account type with the exception of bill payment method • Return to Account type menu

MISCALLENEOUS METHODS:

MODULES	FUNCTIONALITIES
1.WithDraw Money	• Allows user to withdraw a certain amount • Various checks to ensure user if withdrawal Is possible • Exception handling to handle invalid inputs • Updates balance
2.Deposit Money	• Allows user to deposit a certain amount • Exception handling to handle invalid inputs • Method to let the user know if deposition of money was successful • Updates balance
3.Transfer Money	• Allows user to withdraw a certain amount to certain bank and recipient • Various checks to ensure user if Transfer Is possible • Exception handling to handle invalid inputs • Updates balance
4.Zakat Calculation	• Calculates Zakat on the basis of nisab ,compares with base money and updates the balance accordingly • Also lets the user know if Zakat is applicable or not
5.Interest Application	• Calculates annual and monthly interest on basemoney
6.Loan Service	• Allows user to apply for different types of loan in certain ranges • Invokes approve Loan Method to lets user know if loan was approved from the admin panel

BMS REPORT

	• Types of Loans include: • Business Expansion Loan • Mortgage Loan • Credit Builder Loan • Personal Loan
7.Bills Management	• Allows users to handles three types of bills: • Electricity • Gas • Internet • After calculation ,updates the balance accordingly • Lets user know in case of successful payment • Handles various possible Exceptions
8.Transaction History	• Allows user to view their transaction logs including: • Time • Account number • Type • Amount • Information is read from log transaction file
9.Write Complaints	• Allows users to enter their experience with services of bank and make complaints • This information get stored in a separate complaints file present in the admin console
10. Validity Check for Customer console	• This method compares user entered ID and PIN with credentials present in customer file to validate if the user is a true customer of bank

BMS REPORT

	• File Reading implemented
11.Customer File Updating	• Update user entered credentials from new account method to file containing customer information • File Writing implemented
12.Add New Customer	• Assigns a specific id to customer after credentials are entered • Updates the file with the new id to complete new account setup module
13.Account Validity Check	• Validity check to ensure no two customers have same account credentials • File reading implemented • Various Exceptions used
14.Customer Services	• Allow user to view the services provided by banking System before new account setup

Outputs:

MAIN MENU:

```
=====
| WELCOME TO BANK MANAGEMENT SYSTEM |
=====
| MAIN MENU |
|-----|
| 1. Customer Portal |
| 2. Admin Login |
| 0. Exit |
|-----|
=====
Enter your choice: 2
Enter Admin Password: abcde
Access Granted.
```

ADMIN CONSOLE:

1) Customer Records:

```
=====
| ADMIN OPERATIONS CONSOLE |
=====
| 1. Customer Records Management |
| 2. Loan Supervision |
| 3. Credit Card Controls |
| 4. Complaint Resolution Desk |
| 5. Interest & Fee Manager |
| 6. Transaction History Review |
| 7. Account Type setups |
| 0. Logout / Return to Main Gateway |
=====
Select an operation: 1
1,Muhammad umer,muhammad1@example.com,1234",
2,asif jan,Asghar2@example.com,5678",
3,Samina asif,Samina3@example.com,9012",
4,Irfan junejo,irfan4@example.com,3456",
```

2) Loan Approval or Rejection

```
=====
| ADMIN OPERATIONS CONSOLE |
=====
| 1. Customer Records Management |
| 2. Loan Supervision |
| 3. Credit Card Controls |
| 4. Complaint Resolution Desk |
| 5. Interest & Fee Manager |
| 6. Transaction History Review |
| 7. Account Type setups |
| 0. Logout / Return to Main Gateway |
=====
Select an operation: 2
Enter Loan ID to review (e.g., LN-1234): 4573
Loan ID: 4573
Approve this loan? (yes/no): yes
Loan Approved.
```

3) Credit Card Controls:

```
=====
| ADMIN OPERATIONS CONSOLE |
=====
| 1. Customer Records Management |
| 2. Loan Supervision |
| 3. Credit Card Controls |
| 4. Complaint Resolution Desk |
| 5. Interest & Fee Manager |
| 6. Transaction History Review |
| 7. Account Type setups |
| 0. Logout / Return to Main Gateway |
=====
Select an operation: 3
+===== CREDIT CARD CONTROLS =====+
| 1. Issue New Credit Card |
| 2. Block a Credit Card |
| 3. Unblock a Credit Card |
| 0. Back to Admin Panel |
+=====+
Select an option: 1
Enter Customer Account Number to Issue Card: 4689
Credit Card issued for account: 4689
```

```

===== CREDIT CARD CONTROLS =====
|1. Issue New Credit Card      |
|2. Block a Credit Card       |
|3. Unblock a Credit Card     |
|0. Back to Admin Panel       |
=====
Select an option: 3
Enter Credit Card Number to Unblock: 4778
Card 4778 has been UNBLOCKED.

```

```

===== CREDIT CARD CONTROLS =====
|1. Issue New Credit Card      |
|2. Block a Credit Card       |
|3. Unblock a Credit Card     |
|0. Back to Admin Panel       |
=====
Select an option: 0
Returning to Admin Panel...

```

```

===== CREDIT CARD CONTROLS =====
|1. Issue New Credit Card      |
|2. Block a Credit Card       |
|3. Unblock a Credit Card     |
|0. Back to Admin Panel       |
=====
Select an option: 3
Enter Credit Card Number to Unblock: 4778
Card 4778 has been UNBLOCKED.

```

```

===== CREDIT CARD CONTROLS =====
|1. Issue New Credit Card      |
|2. Block a Credit Card       |
|3. Unblock a Credit Card     |
|0. Back to Admin Panel       |
=====
Select an option: 0
Returning to Admin Panel...

```

4) Complaint Desk

```

=====
|          ADMIN OPERATIONS CONSOLE          |
=====
| 1. Customer Records Management |
| 2. Loan Supervision           |
| 3. Credit Card Controls       |
| 4. Complaint Resolution Desk  |
| 5. Interest & Fee Manager     |
| 6. Transaction History Review |
| 7. Account Type setups       |
| 0. Logout / Return to Main Gateway |
=====
Select an operation: 4

The complaints are as follows:
We faced the issue of slow transactions ~ Ali
-----
Mobile app of this bank needs to get fixed ~ Sara
-----
My account balance is not being update ~ Hamza
-----

```

5) Interest and Service Fee Managing

```
=====
|                               |
|      ADMIN OPERATIONS CONSOLE      |
|                               |
| 1. Customer Records Management    |
| 2. Loan Supervision              |
| 3. Credit Card Controls          |
| 4. Complaint Resolution Desk     |
| 5. Interest & Fee Manager        |
| 6. Transaction History Review    |
| 7. Account Type setups          |
| 0. Logout / Return to Main Gateway |
|                               |
|=====|
Select an operation: 5
```

```
-----
INTEREST & SERVICE FEE MANAGER
-----
Enter Account Number: 2355
Enter Principal Amount (PKR): 23445
Annual Interest Rate (%): 5
Time Period (Years): 5
Fixed Service Fee (PKR): 2500

Add another record? (yes/no): yes
```

```
-----
INTEREST & SERVICE FEE MANAGER
-----
Enter Account Number: 9800
Enter Principal Amount (PKR): 769800
Annual Interest Rate (%): 3
Time Period (Years): 3
Fixed Service Fee (PKR): 2500

Add another record? (yes/no): no
```

```
----- INTEREST LOG -----
Account: 2355 | Interest: PKR 6658.98 | Amount: PKR 30103.98 | Fee: PKR 2500.00 | Total: PKR 32603.98
Account: 9800 | Interest: PKR 72495.36 | Amount: PKR 842295.36 | Fee: PKR 2500.00 | Total: PKR 844795.36
-----
```

6) Transaction History

```
=====
|                               ADMIN OPERATIONS CONSOLE                               |
=====
| 1. Customer Records Management |
| 2. Loan Supervision           |
| 3. Credit Card Controls       |
| 4. Complaint Resolution Desk  |
| 5. Interest & Fee Manager     |
| 6. Transaction History Review |
| 7. Account Type setups       |
| 0. Logout / Return to Main Gateway |
=====
Select an operation: 6
2025-06-11 14:36:49 | Account: 12345 | Deposit | $4500.89
2025-06-12 14:37:29 | Account: 17889 | Withdrawal | $4500.89
2025-06-14 20:55:06 | Account: 32989 | Transfer | $23444.0
2025-06-15 08:40:47 | Account: 43148 | Deposit | $12.0
2025-06-15 08:41:58 | Account: 43148 | Bill Payment | $32342.0
2025-06-15 08:42:15 | Account: 12345 | Transfer | $15788.0
```

7) Account Type Setups

```
=====
|                               ADMIN OPERATIONS CONSOLE                               |
=====
| 1. Customer Records Management |
| 2. Loan Supervision           |
| 3. Credit Card Controls       |
| 4. Complaint Resolution Desk  |
| 5. Interest & Fee Manager     |
| 6. Transaction History Review |
| 7. Account Type setups       |
| 0. Logout / Return to Main Gateway |
=====
Select an operation: 7
```

```

=====
|                ACCOUNT TYPE SETUP                |
=====
|  1. View Account Type List                       |
|  2. View Account Type Details                     |
|  0. Back to Admin Menu                           |
=====
Select an option: 2
Account Type Details:
- Savings      | Min Balance: $1000 | Interest: 5%
- Personal     | Min Balance: $800  | Interest: 2.0%
- Salary       | No Min Balance    | Interest: 1.0%
=====
|                ACCOUNT TYPE SETUP                |
=====
|  1. View Account Type List                       |
|  2. View Account Type Details                     |
|  0. Back to Admin Menu                           |
=====
Select an option: 0
Returning to Admin Menu...

```

CUSTOMER CONSOLE:

1) Customer Panel

```

=====
|                WELCOME TO BANK MANAGEMENT SYSTEM                |
=====
|                MAIN MENU                |
=====
|  1. Customer Portal                       |
|  2. Admin Login                           |
|  0. Exit                                 |
=====
Enter your choice: 1
+=====WELCOME TO CUSTOMER PANEL=====+
+-----ACCOUNT STATUS-----+
|1.Log into existing account|
|2.Create a new account    |
|0.Exit                    |
+=====+
Enter your choice: 1
Enter your Account ID: 1
Enter your PIN: 1234
Login successful!

```

2) Exiting account to Account Types menu

```

=====
+=====ACCOUNT TYPES=====+
|1. Savings Account          |
|2. Personal Account         |
|3. Corporate Salary Account |
|4. Exit (Return to Customer Panel)|
+=====+
Enter your choice: 1

```

3) Savings Account Interest

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint            |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 1
Enter time period for which you want to get interest (years): 3
Interest on: 10000000000 PKR, for 3 years at 5.0% = 1.5E8 PKR
Interest generated per month is: 1.25E7

```

4) Withdraw Money

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint            |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 2
Enter amount to withdraw: 12000
Amount 12000.0 withdrawn successfully. Your remaining balance is: 999988000
Transaction logged successfully.

```

5) Deposit Money

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint            |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 3
Enter amount to deposit: 19000
Amount 19000.0 deposited successfully. Your bank balance is: 1000007000
Transaction logged successfully.

```


6) Transferring Money

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint           |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 4
Enter recipient name: Sana
Enter bank name: HBL
Enter amount to transfer: 35000
Transferred Rs. 35000.0 to Sana at HBL.
Remaining balance: Rs. 999972000
Transaction logged successfully.

```

7) Applying for Loan

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint           |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 5
+===== Apply for a Loan =====+
|1. Business Expansion Loan    | ($5,000 - $50,000) |
|2. Mortgage Loan             | ($20,000 - $500,000) |
|3. Credit Builder Loan       | ($500 - $2,000) |
|4. Personal Loan             | ($1,000 - $15,000) |
|0. Exit                      |
+=====+
Enter your choice (0-4): 1
Enter desired loan amount in USD: 10000
Loan application submitted for Business Expansion Loan with amount $10000.0

```

8) Viewing Transaction History

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint           |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 6
2025-06-11 14:36:49 | Account: 12345 | Deposit | $4500.89
2025-06-12 14:37:29 | Account: 17889 | Withdrawal | $4500.89
2025-06-14 20:55:06 | Account: 32989 | Transfer | $23444.0
2025-06-15 08:40:47 | Account: 43148 | Deposit | $12.0

```

9) Filing Complaint

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint            |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 7
We apologize for the inconvenience. Please enter your complaint: Late Transactions
Complaint filed successfully.

```

10) Bill Payment

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint            |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 8
Enter bill type (Electricity, Gas, Internet): internet
Enter bill amount: 600
Paid internet bill of Rs. 600.0
Remaining: Rs. 999971400
Transaction logged successfully.

```

11) Zakat Calculation

```

+=====SAVINGS ACCOUNT=====+
|1. Interest on Money          |
|2. Withdraw Money            |
|3. Deposit Money             |
|4. Transfer Money            |
|5. Apply for loan            |
|6. Transaction History       |
|7. File complaint            |
|8. Bill Payment              |
|9. Zakat Calculation         |
|0. Exit                      |
+=====+
Enter your choice: 9
Zakat on: 999971400 PKR = 2.4999285E7 PKR

```

12) Personal Account Menu

```

+=====PERSONAL ACCOUNT=====+
| 1. Check Balance                |
| 2. Withdraw Money              |
| 3. Deposit Money               |
| 4. Transfer Money              |
| 5. Apply for loan              |
| 6. Transaction History         |
| 7. Bill Payment               |
| 8. File complaint              |
| 9. Zakat Calculation           |
| 0. Exit                       |
+=====+
Enter your choice: █

```

13) Corporate Salary Account

```

+=====CORPORATE SALARY ACCOUNT=====+
| 1. Check Balance                |
| 2. Withdraw Money              |
| 3. Zakat Calculation           |
| 4. Transfer Money              |
| 5. Apply for loan              |
| 6. Transaction History         |
| 7. File complaint              |
| 0. Exit                       |
+=====+
Enter your choice: █

```

14) New Account Creation: Viewing Services

```

+=====WELCOME TO CUSTOMER PANEL=====+
|-----ACCOUNT STATUS-----|
| 1.Log into existing account    |
| 2.Create a new account         |
| 0.Exit                        |
+=====+
Enter your choice: 2
+=====NEW ACCOUNT CREATION=====+
Have a look at the services being provided by our Bank:
+=====SERVICES PROVISION=====+
| 1. Deposit Money              |
| 2. Withdraw Money            |
| 3. Transfer Money            |
| 4. Loan Services              |
| 5. Bill Payment              |
| 6. View History              |
| 7. File Complaint            |
| 8. Check Balance             |
| 0. Exit                      |
+=====+
Select a service you wish to gain knowledge about: 1
DEPOSIT: Our Banking System allows you to deposit money to you account conviniently and without much hassle.

```

15) New Account Setup

```
Select a service you wish to gain knowledge about: 0
Choose our bank for best and reliable outcomes...
Enter your details to create a new account.
Enter your ID: 1234
Enter your name: Ali
Enter your email: Ali12@gmail.com
Enter 4-digit PIN: 6789
Choose Account Type:
1. Corporate Salary Account
2. Personal Account
3. Savings Account
3
Account does not exist.
New customer added successfully. Your information:
1234,Ali,Ali12@gmail.com,6789,Savings Account
```

16) Returning to Main Menu and Exiting

```
=====
| WELCOME TO BANK MANAGEMENT SYSTEM |
=====
| MAIN MENU |
|-----|
| 1. Customer Portal |
| 2. Admin Login |
| 0. Exit |
=====
Enter your choice: 0

System terminated. Thank you for using our Bank.
```

APPENDIX:



Main.java

Text files:



transaction_log.txt



customers.txt



complaints.txt